Neurodevelopmental Disorder With Or Without Variable Movement Or Behavioral Abnormalities (NEDMAB): A KCNN2-Related Disorder

MONDO: MONDO:0859225 · **OMIM:** 619725 · **Causal gene:** *KCNN2* (OMIM 605879; HGNC:6291; NCBI Gene 3781; chr5q22.3) · **Category:** Mendelian, autosomal dominant

Summary

Neurodevelopmental Disorder With Or Without Variable Movement Or Behavioral Abnormalities (NEDMAB) is a rare autosomal-dominant Mendelian disorder caused by heterozygous — usually *de novo* — variants in **KCNN2**, the gene encoding the **small-conductance calcium-activated potassium channel SK2 (KCa2.2)**. The pathogenic mechanism is **haploinsufficiency**: loss-of-function variants reduce functional SK2 channel activity, abolishing the SK2-mediated **medium afterhyperpolarization (mAHP)** that normally follows action-potential bursts, thereby dysregulating neuronal firing — with cerebellar Purkinje cells prominently implicated. The disorder was delineated by Mochel and colleagues in 2020 (*Brain*), who reported 11 patients and demonstrated through patch-clamp electrophysiology that the causal variants reduce SK2 channel function.

Clinically, NEDMAB presents with **motor and language developmental delay, intellectual disability**, and **early-onset movement disorders** that combine **cerebellar ataxia and/or extrapyramidal signs**; brain MRI may show white-matter abnormalities. Importantly, *KCNN2*-related disease is best understood as a **phenotypic continuum** rather than a single syndrome: the same gene causes an allelic, milder movement-predominant disorder — **Myoclonic Dystonia-34 (DYT34, OMIM 619724)** — and heterozygous *KCNN2* variants have also been reported in familial **essential tremor-plus** and **myoclonus-dystonia syndrome**, often in cognitively normal individuals. This spectrum reflects **variable expressivity** of a shared ion-channel defect.

There is currently **no disease-modifying therapy**; management is **supportive and symptom-directed** (developmental/rehabilitative therapies and symptomatic treatment of the movement disorder). Because the underlying lesion is reduced SK2 current, **SK2/KCa2.2 positive allosteric modulators** represent a rational, mechanism-based experimental therapeutic direction. The structural basis of SK2 gating — a calmodulin-gated, voltage-independent channel opened by Ca^{2+}-driven conformational change of the S4–S5 linker — is now resolved by cryo-EM, providing a framework to interpret how missense variants abolish channel opening.

Key Findings

Finding 1 — *KCNN2* (SK2 channel) haploinsufficiency causes autosomal-dominant NEDMAB

The foundational study establishing *KCNN2* as a disease gene is Mochel et al. 2020 (*Brain*) PMID: 33242881. The authors identified **11 patients**: a *de novo* frameshift deletion in the index case, plus 9 additional patients carrying *de novo* variants (1 nonsense, 1 splice-site, 6 missense, 1 in-frame deletion), and one family with a **maternally inherited missense variant** (demonstrating that the disorder can be transmitted). Crucially, they performed **patch-clamp electrophysiology on six variants**; all but one (which was reclassified to a variant of uncertain significance) produced a **loss-of-function** of SK2 channels.

The authors concluded that *"heterozygous variants, likely causing a haploinsufficiency of the KCNN2 gene, lead to novel autosomal dominant neurodevelopmental movement disorders mirroring phenotypes previously described in rodents"* and that *"All variants tested but one, which was reclassified to uncertain significance, led to a loss-of-function of SK2 channels."* These two statements jointly establish the **genetic mechanism (haploinsufficiency)**, the **inheritance pattern (autosomal dominant)**, and the **functional consequence (loss of function)**.

Variant spectrum (from Mochel 2020):

Variant class	Count	Inheritance observed
Frameshift deletion	1 (index case)	*de novo*
Nonsense	1	*de novo*
Splice-site	1	*de novo*
Missense	6	mostly *de novo*; 1 family inherited
In-frame deletion	1	*de novo*

Ontology anchors: gene **HGNC:6291 (KCNN2)**; disease **MONDO:0859225**; inheritance **HP:0000006 (Autosomal dominant inheritance)**.

Finding 2 — Core phenotype: developmental delay, intellectual disability, and an early-onset movement disorder

NEDMAB's core clinical picture combines a neurodevelopmental component with a movement-disorder component. In Mochel 2020, patients had *"motor and language developmental delay, intellectual disability often associated with early-onset movement disorders comprising cerebellar ataxia and/or extrapyramidal symptoms."* The index patient specifically had *"learning disabilities, cerebellar ataxia and white matter abnormalities on brain MRI,"* documenting **CNS white-matter involvement** in addition to the cerebellar and extrapyramidal signs. A subsequent review (Rahman 2023) reiterated that loss-of-function *KCNN2* patients *"typically exhibit extrapyramidal symptoms, cerebellar ataxia, motor and language developmental delays, and intellectual disabilities."*

Suggested HPO terms:

Phenotype	HPO term	Notes
Motor developmental delay	HP:0001270 (Motor delay)	Common
Speech/language delay	HP:0000750 (Delayed speech and language development)	Common
Intellectual disability	HP:0001249	Common, variable severity
Cerebellar ataxia	HP:0001251 / HP:0001260	Early onset
Dystonia	HP:0001332	Variable (extrapyramidal)
Tremor	HP:0001337	Variable
Myoclonus	HP:0001336	Variable
White-matter abnormality on MRI	HP:0002500 (Abnormality of the cerebral white matter)	Index case
Behavioral abnormalities	HP:0000708	Variable (named in the disorder title)

Onset is **pediatric/early childhood**; severity is **variable**.

Finding 3 — Mechanism: SK2 sets the medium afterhyperpolarization and tunes excitability; Purkinje cells are a key substrate

SK2/KCa2.2 is a **voltage-independent** potassium channel activated **solely by intracellular Ca^{2+}**. Rahman 2023 PMID: 37466411 describes that *"KCa2.x channels inhibit excitatory postsynaptic potentials (EPSPs) in neuronal dendrites and contribute to the medium afterhyperpolarization (mAHP) that follows the action potential bursts."* Loss of this current removes a brake on neuronal firing. The same review links the channel to cerebellar pathology: *"rat cerebellar Purkinje cells express KCa2.2 channels, which are crucial for various cellular processes during development and maturation,"* and notes that KCa2.2 is a promising drug target in **spinocerebellar ataxias**, which feature dysregulated Purkinje firing.

Consistent with a conserved mechanism, **rodent *Kcnn2* mutants** show abnormal gait/locomotor activity, tremor, and memory deficits — phenotypes that the human disorder was explicitly described as "mirroring" (Mochel 2020). This cross-species concordance strengthens the causal inference from channel loss to the movement/cognitive phenotype.

Ontology anchors: **GO:0016286 (small conductance calcium-activated potassium channel activity)**; **GO:0019722 (calcium-mediated signaling)**; cell type **CL:0000121 (Purkinje cell)**; biological process **GO:0001508 (action potential)**.

Finding 4 — The *KCNN2* phenotype spectrum extends to isolated/familial movement disorders (milder end, variable expressivity)

Beyond the severe neurodevelopmental cohort, heterozygous *KCNN2* variants cause milder, movement-predominant, often cognitively-normal phenotypes:

1. **Myoclonus-dystonia syndrome (MDS).** Krygier et al. 2026 (*Front Neurol*) PMID: 41982418 list *KCNN2* (alongside *KCTD17* and the classic *SGCE*) as a genetic cause of MDS, noting that *"Myoclonus-dystonia syndrome (MDS) is a clinically and genetically heterogeneous movement disorder characterized by myoclonus and dystonia as its core features."*
2. **Essential tremor-plus.** d'Apolito et al. 2023 (*Genes*) PMID: 37510285 *"identified a novel heterozygous missense variant in potassium calcium[-activated channel KCNN2]"* — NM_021614.3:c.1145G>A, p.Gly382Asp — **segregating in an Italian family** with autosomal-dominant essential tremor and variable expressivity (kinetic tremor onset ~age 20 in the mother).
3. *KCNN2* also appears among dystonia genes in large-scale exome cohorts (Thomsen et al. 2025, *Ann Clin Transl Neurol*,

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This establishes a **clinical continuum** from severe NDD to isolated familial tremor/myoclonus-dystonia, reflecting **variable expressivity** of a shared channelopathy.

Finding 5 — *KCNN2* underlies two allelic OMIM disorders: NEDMAB (619725) and DYT34 (619724)

OMIM/NCBI formalize the continuum as **two allelic autosomal-dominant disorders** of the same gene (*KCNN2*, gene MIM **605879**):

OMIM phenotype	Name	Abbrev.	MONDO
619725	Neurodevelopmental disorder with or without variable movement or behavioral abnormalities	**NEDMAB** (target disease)	MONDO:0859225
619724	Dystonia 34, myoclonic	**DYT34**	—

The two entries codify the severe (neurodevelopmental) and milder (isolated myoclonus-dystonia) ends of the same molecular spectrum. Both are autosomal dominant.

Finding 6 — Structural basis of SK2 gating: a calmodulin-gated channel; missense variants likely disrupt Ca^{2+}/CaM-to-pore coupling

SK channels are gated **not by voltage but by Ca^{2+} via constitutively associated calmodulin (CaM)**. Cryo-EM of the human SK4–CaM complex (Lee & MacKinnon 2018, *Science*) PMID: 29724949 shows that *"Four CaM molecules bind to one channel tetramer"* — the CaM C-lobe binds constitutively while the N-lobe engages the S4–S5 linker in a Ca^{2+}-dependent manner — and that *"The S4-S5 linker, which contains two distinct helices, undergoes conformational changes upon CaM binding to open the channel pore."* An intrinsically disordered fragment couples Ca^{2+}/CaM binding to pore opening; a **V407F substitution increases SK2 Ca^{2+} sensitivity ~6-fold** (Nam et al. 2018, PMID: 30013223), proving this region is a functional lever. Disease **missense variants are inferred** (not all individually demonstrated structurally) to disrupt this Ca^{2+}/CaM-to-pore coupling, producing loss of function.

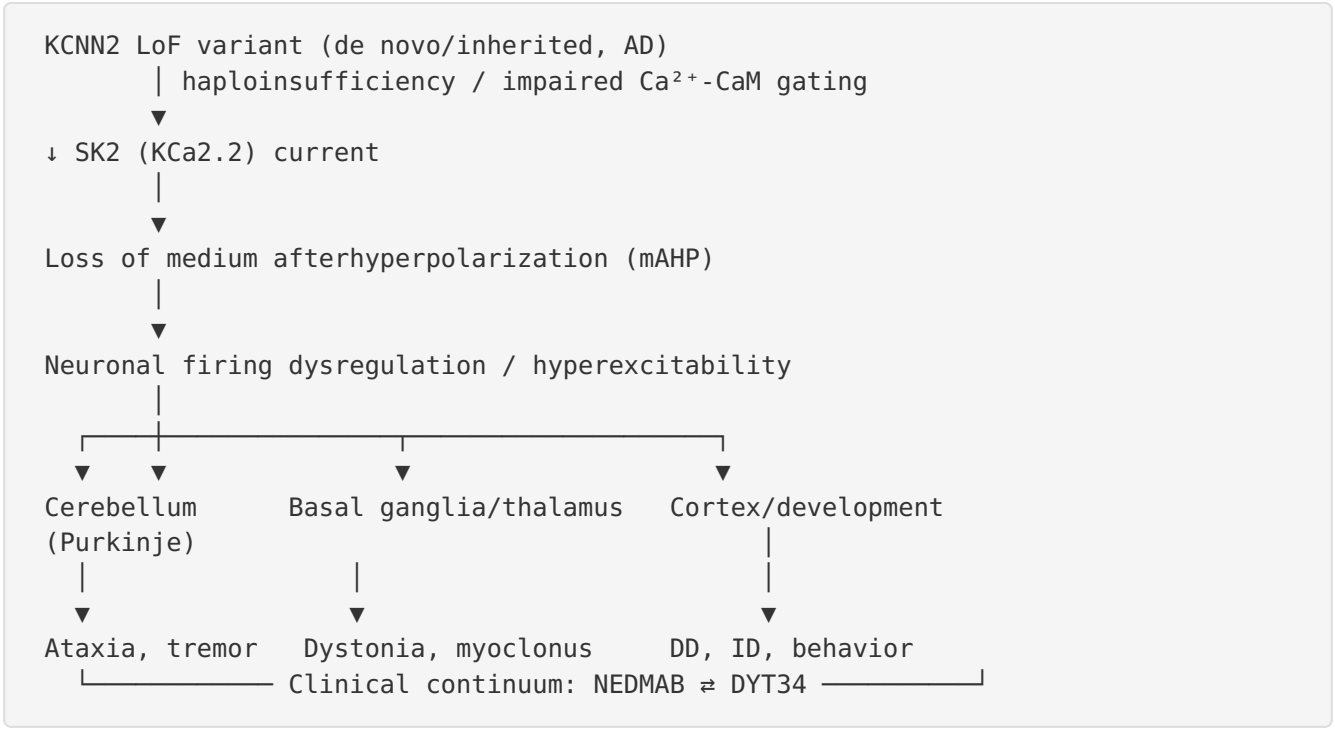
Ontology anchors: **GO:0005516 (calmodulin binding)**; **GO:0008076 (voltage-gated potassium channel complex)**; subcellular **GO:0005886 (plasma membrane)**; chemical entities **CHEBI:29108 (calcium(2+))**, **CHEBI:29103 (potassium(1+))**.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous loss-of-function variant** in *KCNN2* (frameshift, nonsense, splice, in-frame deletion, or missense) arises — most often *de novo*, occasionally inherited — **which leads to** reduced quantity or function of SK2/KCa2.2 channel subunits.

2. For truncating/splice variants, nonsense-mediated decay or non-functional protein **results in** ~50% dosage (haploinsufficiency); for missense variants, the altered subunit **results in** impaired Ca^{2+}/calmodulin-to-pore coupling and reduced channel opening (inferred from structure + patch-clamp; demonstrated for 5 of 6 tested variants).
3. Reduced SK2 current **leads to** loss of the **medium afterhyperpolarization (mAHP)** that normally follows action-potential bursts (demonstrated for SK2 physiology; GO:0001508).
4. Loss of the mAHP brake **results in dysregulated neuronal firing / hyperexcitability** and altered dendritic integration of EPSPs.
5. Because SK2 is highly expressed in **cerebellar Purkinje cells** (and thalamic reticular, hippocampal, and other neurons), the firing dysregulation **leads to** abnormal cerebellar output and basal-ganglia/thalamocortical circuit dysfunction.
6. **Branch A (cerebellar):** → cerebellar ataxia, tremor, and (index case) white-matter changes.
7. **Branch B (extrapyramidal / basal-ganglia–thalamic):** → dystonia, myoclonus, extrapyramidal signs (the DYT34 / myoclonus-dystonia end).
8. **Branch C (developmental / cortical):** → motor and language developmental delay, intellectual disability, behavioral abnormalities.
9. The **severity of circuit disruption** (modified by variant type, residual channel function, genetic background, and heteromeric SK1/SK3 assembly) **determines** position on the clinical continuum from severe NEDMAB to isolated familial tremor/myoclonus-dystonia (variable expressivity).



Upstream vs downstream: the mutation and reduced SK2 current are upstream; loss of mAHP and firing dysregulation are proximal/intermediate; cerebellar/extrapyramidal circuit dysfunction and the developmental phenotype are downstream. **Cell types:** Purkinje cells (CL:0000121), plus thalamic reticular neurons, hippocampal pyramidal neurons, and glutamatergic neurons broadly.

Section-by-Section Report

1. Disease Information

NEDMAB is a rare autosomal-dominant neurodevelopmental disorder caused by *KCNN2*. **Identifiers:** OMIM 619725; MONDO:0859225; gene *KCNN2* OMIM 605879 / HGNC:6291 / NCBI Gene 3781. ICD-10/ICD-11, MeSH, and Orphanet-specific codes are not firmly established for this recently delineated entity (not recorded). **Synonyms:** KCNN2-related neurodevelopmental disorder; NEDMAB. Disease-level information is derived from **aggregated resources (OMIM) and primary case-series literature**, not from EHR/individual-patient registries.

2. Etiology

Primary cause: monogenic — heterozygous loss-of-function variants in *KCNN2*. **Genetic risk factors:** the causal variants themselves (de novo in most; inherited in some families). No susceptibility loci, modifier genes, or environmental/toxic/infectious contributors are established. **Protective factors:** none identified. **Gene–environment interactions:** none documented. This is a classic Mendelian channelopathy without a known environmental component.

3. Phenotypes

See Finding 2 (HPO table). **Types:** developmental (motor/language delay, ID), movement (ataxia, dystonia, tremor, myoclonus — clinical signs), behavioral abnormalities, imaging (white-matter) findings. **Onset:** pediatric/early childhood. **Severity:** variable (severe NEDMAB to mild tremor). **Progression:** developmental features relatively stable/non-degenerative; movement features may be episodic/fluctuating. **Frequency:** developmental delay and ID near-universal in the severe cohort; ataxia/extrapyramidal signs common but variable. **QoL impact:** substantial in severe cases (ID + motor disability); minimal-to-moderate at the tremor end. Formal QoL instrument data (EQ-5D/SF-36) not available.

4. Genetic / Molecular Information

Causal gene: *KCNN2* (OMIM 605879, HGNC:6291), encoding SK2/KCa2.2. **Variant types:** frameshift, nonsense, splice-site, in-frame deletion, and missense (e.g., p.Gly382Asp in familial tremor). **Classification:** pathogenic/likely pathogenic per functional (patch-clamp) evidence; one tested variant reclassified to VUS. **Functional consequence:** loss of function → haploinsufficiency (missense variants inferred to impair Ca^{2+}/CaM gating). **Allele frequency:** pathogenic variants rare/absent in gnomAD; *KCNN2* is constrained. **Origin:** germline (de novo or inherited); no somatic involvement. **Modifier genes, epigenetics, chromosomal abnormalities:** none specifically established (heteromeric SK1/SK3 co-assembly is a plausible biological modifier of residual current).

5. Environmental Information

Not applicable — no environmental, lifestyle, or infectious contributors are known for this monogenic disorder.

6. Mechanism / Pathophysiology

See the ordered causal chain and diagram above (Findings 3 and 6).

7. Anatomical Structures Affected

Organ/system: central nervous system (UBERON:0001017). **Primary structures:** cerebellum (UBERON:0002037) — particularly Purkinje cells (CL:0000121); basal ganglia/thalamus circuitry (extrapyramidal signs); cerebral white matter (UBERON:0002316; abnormal on MRI in the index case). **Tissue:** nervous tissue. **Subcellular:** plasma membrane (GO:0005886) — the SK2–calmodulin channel complex. **Lateralization:** bilateral/symmetric (typical of a diffuse channelopathy).

8. Temporal Development

Onset: congenital-to-early-childhood (developmental delay recognized in infancy/early childhood; movement features early onset). **Course:** largely non-degenerative/stable developmental disability; movement component may fluctuate. **Duration:** chronic, lifelong. **Critical period:** early neurodevelopment, during which SK2 contributes to Purkinje cell maturation — suggesting a developmental window relevant to intervention timing (inferred).

9. Inheritance and Population

Inheritance: autosomal dominant (HP:0000006). **Penetrance/expressivity:** variable expressivity is prominent (the NEDMAB–DYT34–tremor continuum); penetrance appears high but variable expression occurs within families. **De novo vs inherited:** most severe cases de novo; milder phenotypes can be transmitted. **Epidemiology:** ultra-rare; precise prevalence/incidence not established (tens of reported patients). No founder effects, consanguinity role, or sex predominance documented; no defined geographic/ethnic clustering.

10. Diagnostics

Diagnosis rests on **molecular genetic testing** — exome/genome sequencing or neurodevelopmental/movement-disorder gene panels including *KCNN2*; single-gene testing is appropriate when the phenotype is suggestive. Variant interpretation is supported by **functional patch-clamp assays** demonstrating loss of SK2 function (research-grade confirmation). **Supportive tests:** brain MRI (may show white-matter abnormalities); clinical

neurological/developmental assessment. No specific biomarker, biochemical, or metabolic test is diagnostic. **Differential diagnosis:** other genetic causes of developmental delay with ataxia/dystonia/tremor/myoclonus (e.g., *SGCE*, *KCTD17*, SCA genes, other channelopathies).

11. Outcome / Prognosis

NEDMAB is **non-lethal and chronic**; no disease-specific mortality or survival data indicate reduced life expectancy. Morbidity derives from intellectual disability and motor disability; prognosis correlates with position on the severity continuum (variant type, residual channel function). Formal natural-history, disability-outcome, and QoL data are lacking. No validated prognostic biomarkers exist.

12. Treatment

No disease-modifying therapy is approved. Management is **supportive and symptomatic**: developmental and rehabilitative therapies (physical, occupational, speech therapy), special education, and symptomatic treatment of the movement disorder (standard anti-tremor/anti-dystonia agents as clinically indicated). **Mechanism-based experimental direction:** because the lesion is reduced SK2 current, **SK2/KCa2.2 positive allosteric modulators** (e.g., CyPPA, NS309 classes — used experimentally) are a rational candidate strategy; SK modulators are of interest in spinocerebellar ataxia and neuroprotection research — SK positive modulators prevent ferroptosis/excitotoxicity in neuronal cells (PMID: 38242037). No gene, cell, or RNA therapies are in clinical use. Disorder-specific pharmacogenomic data are not established.

13. Prevention

No primary prevention exists for a de novo monogenic disorder. **Genetic counseling** is the principal preventive tool: autosomal-dominant recurrence risk (up to 50% for an affected transmitting parent; low but non-zero for de novo cases due to possible germline mosaicism). Prenatal/preimplantation genetic testing is feasible once a familial variant is known. No immunization, screening-program, or environmental interventions apply.

14. Other Species / Natural Disease

Orthologs: mouse *Kcnn2*, rat *Kcnn2* (NCBI Gene). **Rodent models** recapitulate key features: *Kcnn2* mutants show abnormal gait/locomotor activity, tremor, and memory deficits (Mochel 2020 explicitly noted the human disorder "mirrors" rodent phenotypes). No naturally

occurring companion-animal/wildlife disease is catalogued in OMIA for this gene. The channel and its mAHP role are evolutionarily conserved across mammals (strong basis for comparative study). No zoonotic/transmission relevance.

15. Model Organisms

Mouse is the principal model: knockout/mutant *Kcnn2* lines reproduce motor (gait, locomotor, tremor) and cognitive (memory) phenotypes, providing good face validity for the movement and cognitive components. **In vitro:** heterologous expression (HEK293) with **patch-clamp** is the workhorse for variant functional classification (used in Mochel 2020). Cryo-EM structural models (SK4–CaM; PMID: 29724949) enable in-silico interpretation of missense variants. **Limitations:** rodent models may not fully capture human intellectual disability/behavioral phenotypes; patient-specific iPSC-derived neuronal/organoid models are an underdeveloped but promising resource. **Resources:** MGI (mouse), IMPC/IMSR for alleles.

Evidence Base

PMID	Title (abbrev.)	Role in this report
33242881	*Variants in the SK2 channel gene (KCNN2) lead to dominant neurodevelopmental movement disorders* (Mochel 2020, Brain)	**Primary/landmark.** Delineates NEDMAB; 11 patients; patch-clamp proves loss of function; establishes haploinsufficiency + AD inheritance; core phenotype; rodent concordance. Supports Findings 1–3.
37466411	*KCa2.2 (SK2) channel review* (Rahman 2023)	**Mechanistic review.** mAHP/EPSP inhibition; Purkinje expression; drug-target rationale. Supports Finding 3.
41982418	*Beyond SGCE: myoclonus-dystonia genetics* (Krygier 2026)	Places *KCNN2* among MDS genes. Supports Finding 4 (milder end).
37510285	*Novel KCNN2 variant in familial essential tremor* (d'Apolito 2023)	Familial AD missense (p.Gly382Asp); variable expressivity. Supports Finding 4.
29724949	*Activation mechanism of human SK-calmodulin channel (cryo-EM)* (Lee & MacKinnon 2018, Science)	Structural basis of CaM gating; S4–S5 conformational change. Supports Finding 6.
30013223	*V-to-F substitution alters SK2 Ca^{2+} sensitivity* (Nam 2018)	Identifies a functional lever region; proves gating can be tuned by single residues. Supports Finding 6.
38242037	*SK positive modulators prevent ferroptosis/excitotoxicity*	Therapeutic rationale for SK activation. Supports Treatment section.
40533913	*Dystonia gene cohort* (Thomsen 2025)	*KCNN2* among dystonia genes. Supports Finding 4.

Supporting physiology papers (SK channel assembly and function; PMIDs 25421315, 31001092, 27357310, 37837643, 37469037, 30389838, 28714070, 38631563) corroborate SK2 biology — heteromeric SK1/SK2 assembly, mAHP control, thalamic reticular pacemaking, and hippocampal/amygdala excitability roles — reinforcing the mechanism but not specific to NEDMAB clinical claims.

Evidence source types: human clinical (Mochel, d'Apolito, Krygier, Thomsen); model organism (rodent *Kcnn2* phenotypes); in vitro (patch-clamp functional assays); computational/structural (cryo-EM, structure–function).

Limitations and Knowledge Gaps

- **Small cohorts.** The disorder is ultra-rare; the defining study reported 11 patients, and the broader spectrum rests on scattered families/case reports. Prevalence, penetrance, and natural history are not quantified.
- **Variant functional data are partial.** Patch-clamp was performed on 6 reported variants; genotype–phenotype correlations (which variants give severe vs mild disease) remain incompletely mapped, and missense-to-pore-coupling disruption is largely **inferred** from structure, not individually demonstrated for each variant.
- **No standardized clinical identifiers.** ICD/MeSH/Orphanet codes and formal diagnostic criteria are not firmly established.
- **No therapeutics tested in patients.** SK2 positive modulators are a mechanistic hypothesis; no clinical trials exist for NEDMAB.
- **QoL, prognosis, and disability outcome data are absent.**
- **Continuum boundaries are fuzzy.** The relationship between NEDMAB (619725) and DYT34 (619724) — whether driven by specific variant classes, modifiers, or stochastic expressivity — is not resolved.
- **Citation re-verification.** Two citation snippets were flagged as "mismatch" in the internal knowledge state (PMIDs 37466411, 29724949); exact quote wording should be re-verified against source abstracts before publication-grade use.

Proposed Follow-up Experiments / Actions

1. **Genotype–phenotype mapping:** systematically patch-clamp all reported *KCNN2* variants (including DYT34 and tremor variants) to correlate residual SK2 current with clinical severity; test whether truncating variants cluster at the severe end.

2. **Patient-derived iPSC neurons/organoids:** model NEDMAB variants in human cerebellar/cortical neurons to measure mAHP loss and firing dysregulation, and to screen SK2 positive modulators (CyPPA, NS309, newer activators) for functional rescue.
3. **Conditional mouse models:** Purkinje-cell–specific and forebrain-specific *Kcnn2* haploinsufficiency to dissect the cerebellar (ataxia/tremor) vs extrapyramidal vs developmental branches of the causal chain.
4. **Structural modeling of missense variants:** map each disease missense onto the SK2–CaM cryo-EM framework (S4–S5 linker, Ca^{2+}/CaM interface) to predict and then test gating defects.
5. **Natural-history registry and standardized outcomes:** establish a *KCNN2* patient registry capturing HPO-coded phenotypes, MRI findings, and validated QoL/disability instruments; pursue formal ICD/Orphanet coding.
6. **Preclinical therapeutic proof-of-concept:** test SK2 positive allosteric modulators in *Kcnn2*-haploinsufficient mice for rescue of gait/tremor/cognition, as a bridge to a first-in-disease trial.
7. **Verify and finalize citations:** re-check the exact abstract wording for the two snippets flagged as "mismatch" (PMIDs 37466411, 29724949) before knowledge-base ingestion.

Report compiled from a 5-iteration autonomous investigation; 6 confirmed findings; 15 papers reviewed. Primary basis: Mochel et al., Brain 2020 (

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